

CASE STUDY 4: MULTI-LEVEL ESTATE WIRELESS OVERHAUL

Enterprise-Grade Wi-Fi Mesh Architecture, Master-Class Cat6a Cabling & Dead Zone Elimination

PRIMARY DOMAIN High-Density Residential RF Engineering & Structured Cabling	ESTATE LOCATION Mission Hills, KS (ZIP 66208)	HARDWARE STACK Wi-Fi 6E Tri-Band APs, PoE+ Managed Switches, Cat6a Shielded Cable
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I. PROJECT OVERVIEW & RESIDENTIAL CHALLENGE

Client Profile & Architectural Silos: A sprawling 12,000+ square foot luxury residential estate in Mission Hills, Kansas (66208), owned by a highly visible and influential Kansas City resident and beloved civic hero. The property featured a full subterranean concrete basement, three main above-ground living levels, and extensive outdoor grounds extending all the way to the pool deck. The structure incorporated thick lath-and-plaster walls, reinforced concrete subfloors, and large dual-pane Low-E glass sliding doors that created heavy RF barriers. The estate housed over 85 smart home devices, networked televisions across every room and outdoor pool cabana, high-end security cameras, and automated lighting systems.

The Catalyst (Chronic Connectivity Failures): The household suffered from total wireless failure across nearly 70% of the estate and zero coverage outdoors by the pool. Antiquated consumer-grade wireless extenders and daisy-chained standalone Access Points (APs) created severe channel interference, frequent packet loss, and constant smart TV buffering or video conference dropouts. The subterranean basement, pool deck, and top-floor master suite were completely dead zones.

Operational Impact: Due to frequent high-profile remote engagements, charity coordination calls, and media obligations, reliable connectivity was mission-critical. The homeowner could not reliably stream 4K video or participate in secure video conferences, security cameras suffered intermittent offline timeouts, and smart home automation routines failed regularly.

II. STAKEHOLDER DISCOVERY & INTERVIEW MATRIX

Stakeholder Role	Primary Concerns & Questions	Uncovered Pain Points	Design Impact & Resolution
Principal Homeowner / KC Civic Hero	- Why do virtual charity meetings and 4K TVs constantly freeze? - Can we get gigabit speeds everywhere from the basement to the pool?	- Complete dead zones in executive suite, basement media room, and pool deck. - Frequent wireless extender reboots during critical broadcasts.	Designed enterprise Wi-Fi 6E mesh with dedicated 6GHz wired backhaul channels and weather-rated outdoor APs.
Estate Manager & Operations	- Why do security cameras and smart locks go offline weekly? - How do we secure guest access for visiting dignitaries?	- Complete lack of outdoor Wi-Fi coverage at the pool deck. - IoT device congestion on main residential VLAN.	Implemented rigorous VLAN segmentation and ruggedized outdoor weather-resistant APs.
Custom Integrator / Low-Voltage Lead	- How do we pull cables through 4 stories of historic masonry and out to the pool? - Where can we mount APs aesthetically?	- Zero existing conduit between basement, top floor, and pool cabana. - Plaster walls blocking RF signals.	Partnered with elite subcontracted cable pullers for core-drilling, trench-conduit routing, and fish-tape execution.
Family Members / Support Team	- Why does streaming buffer in the basement and by the pool? - Why are online gaming ping spikes so high?	- High latency over wireless daisy-chained repeaters. - Co-channel interference.	Decommissioned all wireless repeaters in favor of hardwired PoE ceiling and outdoor APs.

III. FORMAL SCOPE BOUNDARY STATEMENT

IN-SCOPE Deliverables - Comprehensive RF Site Survey and signal attenuation mapping from basement to pool deck. - Structured Cat6a cable pulls by elite subcontracted technicians from central rack to indoor ceiling and outdoor APs. - Deployment of PoE+ managed core switching with VLAN segmentation for all IoT devices and smart TVs. - Installation of enterprise Wi-Fi 6E tri-band indoor and outdoor access points with seamless roaming. - Guest portal and secure isolated IoT VLAN configuration across all 85+ smart devices and televisions.	OUT-OF-SCOPE Boundaries - Audio/Visual receiver programming or zone amplifier configuration. - Smart appliance repair or motorized window shade controller programming. - Carrier ISP fiber-optic circuit speed upgrades (relying on existing 1 Gbps fiber drop). - Landscape lighting or outdoor security floodlight electrical wiring.
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■ IV. PRE-DESIGN RF SITE SURVEY & DIAGNOSTIC ANALYSIS

Conducted an exhaustive on-site RF spectrum analysis using professional site survey software and spectrum analyzers across all four levels and exterior pool grounds. The evaluation mapped severe structural signal attenuation:

[RF Spectrum & Attenuation Diagnostic Summary] 1. Structural & Environmental Attenuation Analysis: |— Reinforced Concrete Subfloors & Earth Retention: 18-24 dB loss (basement Faraday cage) |— Lath, Plaster & Low-E Glass Doors: 15-22 dB attenuation blocking indoor-to-pool propagation |— Outdoor / Pool Area: Complete signal drop-off past exterior masonry walls 2. Legacy Equipment Performance: |— Smart TV & IoT Buffering: High packet loss (28.4%) causing continuous streaming drops on televisions from basement to pool |— Roaming Latency: Client devices hanging onto weak APs for 15+ seconds before handoff

Resolution: Diagnostic data proved that wireless repeating was mathematically incapable of servicing a 12,000 sq. ft. masonry structure and its exterior grounds. The design mandated physical hardwiring: running certified Cat6a cabling through vertical utility chases and weather-sealed conduit to ceiling-mounted enterprise access points indoors and ruggedized weather-resistant APs outside by the pool.

■ V. CHRONOLOGICAL STEP-BY-STEP EXECUTION SEQUENCE

- P1

RF Survey & Heatmapping: Mapped dead zones across basement, living levels, and outdoor pool grounds.
- P2

Structured Cabling & Master-Class Pulls: Deployed elite subcontracted cable pullers to fish shielded Cat6a cables indoors and run weather-sealed lines out to the pool cabana with immaculate precision.
- P3

Core Switch & Rack Reorganization: Installed wall-mount rack, enterprise router, and PoE+ managed switch with UPS backup.
- P4

VLAN Partitioning: Configured dedicated virtual LANs for Primary Residential, IoT Devices, Smart TVs, and Guest Access.
- P5

AP Deployment & Outdoor Tuning: Mounted Wi-Fi 6E tri-band access points indoors and ruggedized weather-rated APs outside by the pool.
- P6

Validation & Verification: Performed post-installation verification heatmaps confirming 100% gigabit coverage from basement to pool.

■ VI. KEY METRICS & PERFORMANCE OUTCOMES

Metric Category	Pre-Migration Baseline	Post-Implementation Outcome
Wi-Fi Coverage Footprint	30% coverage (Basement, pool, & top floors completely dead)	100% Full-Estate Coverage (Basement to outdoor pool deck)
Smart TV & IoT Streaming	Continuous buffering and dropouts on smart TVs	Flawless 4K Streaming across all televisions and IoT devices
IoT & Camera Stability	Frequent weekly offline timeouts (85+ devices)	100% Uptime on isolated IoT VLAN
Roaming & Handoff Latency	15+ seconds drop on moving between floors	Seamless Fast Roaming (<50ms handoff)

■ VII. RETROSPECTIVE REALITY, FINANCIAL TRANSPARENCY & OUTCOME

Transparent Financial Structure & Historic Investment: Initial investment was not a concern; our team maintained complete transparency regarding the financial structure, the age of the historic home, and the detailed breakdown of costs relative to the scope required. Clients appreciated our upfront clarity on budgeting for complex masonry penetrations.

Master-Class Cable Pulling Execution: Bridging the extreme physical divide—from the subterranean concrete basement, through multi-layer lath-and-plaster walls, past heavy Low-E glass sliding doors, and all the way out to the open-air pool deck—required specialized outdoor weather-resistant access points and trench-routed shielded Cat6a cabling. Our subcontracted cable pullers executed this work with extraordinary precision, protecting the home's historic aesthetic while delivering flawless cable certification.

IoT & Smart Television Integration: By integrating all 94+ smart devices, networked televisions, and pool automation systems onto an isolated, high-priority IoT VLAN, we eliminated bandwidth contention and guaranteed buffer-free 4K streaming anywhere on the property.

Final Project Outcome: The transformation was absolute. The homeowner, a celebrated Kansas City resident and civic hero, praised the flawless performance, noting that virtual charity meetings, high-stakes media broadcasts, and family streaming operated with zero interruption across every corner of the estate—from the basement theater to the poolside loungers.